

8. A Super Ball is launched 100 feet into the air from ground level, measured vertically to the bottom of the ball. It falls back to ground level and then rebounds $\frac{3}{4}$ of its height on the previous bounce. The ball continues this pattern until it stops. Determine the total distance, in feet, the ball traveled as measured from ground level vertically to the bottom of the ball.
9. The equation $(y - 2)^2 = 5(x + 3)$ for the standard coordinate system is rewritten in terms of a coordinate system (x', y') for which the new origin is $(5, 2)$ with axes parallel to the original xy -axes. The resulting equation can be written in the form: $(y')^2 = k(x' + w)$ where k and w are positive integers. Determine the value of $(k + w)$.
10. Assume the following is NOT a degenerate form of a conic. Identify $6x^2 + 5xy + 2y^2 + x + 2y - 3 = 0$ as a circle, ellipse, hyperbola, or parabola. Express your answer by writing the whole word for this conic.
11. $i = \sqrt{-1}$. Complex $z = 1 + i\sqrt{3}$. One square root of z is $\sqrt{z} = a + bi$ with a and b exact real numbers and in simplified radical form and with $a > 0$. Determine the ordered pair (a, b) . Express your answer as the ordered pair (a, b) with exact entries.
12. Twins Jordan and Michael have the following growth curves, with x measured in years and y in inches: $y_1 = 20 + 54\left(1 - e^{\left(\frac{-x}{3}\right)}\right)$ and $y_2 = 21 + 53\left(1 - e^{\frac{-x}{3.1}}\right)$. Determine the sum of all the year(s) after birth at which their heights were equal. Express your answer as a decimal rounded to the nearest thousandth.
13. The rectangular coordinates $(-2, -2\sqrt{3})$ can be written in polar form as (r, θ) where $0 \leq \theta < 2\pi$ in radian measure. Determine these polar coordinates. Express your answer as the ordered pair (r, θ) .
14. In $\triangle ABC$, $\angle BAC = 74^\circ$ and $AC = 100$. $\overline{BC}$ has integral length and is shorter than $\overline{AC}$. Determine the sum of all possible integral lengths for $\overline{BC}$.